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[iNSPIRE-HEP](#)

Keywords	Theoretical cosmology; interacting dark sectors; effective field theory; cosmological stability; dynamical systems; modified gravity; CLASS/Cobaya and Bayesian cosmology.	
Research Experience	University College London Department of Mathematics <i>EPSRC Doctoral Prize Postdoctoral Research Fellow</i>	2024 - 2026
Education	University College London <i>Ph.D. Mathematics</i> (awarded December 28th 2023) Thesis: ‘A unified approach to geometric modifications of gravity’ Advisor: Prof. Christian G. Boehmer	2019 - 2023
	University of Nottingham <i>M.Sc. Gravity, Particles and Fields (Distinction)</i> Thesis: ‘Scalar fields in cosmology: a dynamical systems approach’ Advisor: Prof. Edmund J. Copeland	2017 - 2018
	University of Nottingham <i>B.Sc. Physics with Theoretical Physics (Hons)</i> Thesis: ‘Vacuum instability of the Standard Model of Particle Physics’ Advisor: Dr. Peter Millington	2014 - 2017
Publications	Corresponding (lead) author denoted with [†] 12. <i>Non-minimal fluid Lagrangian couplings</i> C. G. Boehmer, E. Jensko[†], E. Al-Nasrallah Phys.Rev.D 114 (2026) 4, 044080 [2605.16220 gr-qc] 11. <i>Interacting dark sector from intrinsic entropy couplings</i> E. Jensko[†], E. M. Teixeira and V. Poulin Phys.Rev.D 114 (2026) 2, 023536 [2603.10622 astro-ph.CO] 10. <i>Cosmological dynamical systems of non-minimally coupled fluids and scalar fields</i> H. Ashi, C. Boehmer, A. del Sordo, E. Jensko[†] Gen.Rel.Grav. 57 (2025) 12, 167 [2509.14440 gr-qc] 9. <i>Geometric formulation of k-essence and late-time acceleration</i> L. Csillag and E. Jensko[†] JCAP 09 (2026) 028 [2505.15975 gr-qc] 8. <i>The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics</i> (contributor) Phys.Dark Univ. 49 (2025) 101965 [2504.01669 astro-ph.CO] 7. <i>Spatial curvature in coincident gauge $f(Q)$ cosmology</i> E. Jensko[†] Class.Quant.Grav. 42 (2025) 5, 055011 [2407.17568 gr-qc]	

6. *Relativistic interacting fluids in cosmology*
D. Iosifidis, **E. Jensko** and T. Koivisto[†]
JCAP 11 (2024) 043 [[2406.01412](#) gr-qc]
5. *A new 2D formulation of modified General Relativity*
C. Boehmer[†] and **E. Jensko**
Fortsch.Phys. 72 (2024) 6, 2300216 [[2308.01728](#) gr-qc]
4. *Dynamical systems analysis of $f(Q)$ gravity*
C. Boehmer, **E. Jensko** and R. Lazkoz[†]
Universe 9 (2023) 4, 166 [[2303.04463](#) gr-qc]
3. *Modified gravity: A unified approach to metric-affine models*
C. Boehmer and **E. Jensko**[†]
J.Math.Phys. 64 (2023) 8, 082505 [[2301.11051](#) gr-qc]
2. *Cosmological dynamical systems in modified gravity*
C. Boehmer, **E. Jensko** and R. Lazkoz[†]
Eur.Phys.J.C 82 (2022) 6, 500 [[2201.09588](#) gr-qc]
1. *Modified gravity: a unified approach*
C. Boehmer[†] and **E. Jensko**
Phys.Rev.D 104 (2021) 2, 024010 [[2103.15906](#) gr-qc]

Research Funding	<i>UCL Global Engagement Funds (GEF)</i> (Co-applicant; £3,900) 2026 <i>Strengthening university research in Bosnia and Herzegovina through mathematical modelling and cosmology</i> University of Tuzla, Bosnia and Herzegovina, July 2026
	<i>International Visiting Scholar & Remote Collaboration Grant</i> (ca. £9,900) 2023 <i>Investigating strong coupling with effective field theory methods</i> University of Science and Technology of China, March–May 2023
Fellowships and Training Grants	<i>EPSRC Doctoral Prize Fellowship extension</i> 2026 Six-month postdoctoral fellowship extension at UCL, April–October 2026
	<i>EPSRC Doctoral Prize Fellowship</i> 2024–2026 Two-year postdoctoral fellowship at UCL, April 2024–April 2026
	<i>EPSRC PhD Studentship</i> 2019–2023 <i>Lorentz invariance and diffeomorphism invariance in modified gravity</i> Fully funded PhD studentship, UCL Department of Mathematics
Travel and Mobility Grants	<i>COST CA24101: Short Term Scientific Mission (STSM)</i> 2026 <i>Thermodynamic dark sector models and phase-space structures for fundamental physics</i> University of Wrocław, Poland, September 2026
	<i>UCL MAPS Early Career Researchers Travel Grant</i> 2026 <i>Interacting dark sector models: cosmology of elastic scattering</i> Universidad Politécnica de Madrid, Spain, June 2026
	<i>Institute of Physics Early Career Researchers Fund</i> 2025 Annecy-le-Vieux Laboratory of Theoretical Physics (LAPTh), France, December 2025
	<i>COST CA21136: Short Term Scientific Mission (STSM)</i> 2025 <i>Investigating cosmic tensions with generalised interacting dark sector models</i>

Universidad de Salamanca, Spain, October 2025

COST CA21136: Short Term Scientific Mission (STSM) 2024
Cosmological dynamics in geometrical gravitational theories
Babeş-Bolyai University, Romania, October 2024

London Mathematical Society Travel Grant for Early Career Researchers 2023

Dora Plus Grant for Visiting Doctoral Students 2021
University of Tartu, Estonia, August–October 2021

Awards

IOP Publishing Impact Award 2025
Selected for a collection celebrating standout UK research for my work on [spatial curvature in \$f\(Q\)\$ cosmology](#)

David Warren Fund Scholarship 2022
For outstanding work in Graduate Studies at University College London

Andrew Rosen Prize in Applied Mathematics 2021
For outstanding work in Graduate Studies at University College London

Research Visits

University of Science and Technology of China November–December 2026
Department of Astronomy / Particle Cosmology Group
Host: Prof. Yi-Fu Cai
One-month invited research visit, working on extended dark matter models.

University of Montpellier November 2025
Laboratoire Univers et Particules de Montpellier
Host: Dr. Vivian Poulin
Collaboration on entropy coupled interacting dark sector models.

Universidad de Salamanca October 2025
IUFFyM / Gravitation and Cosmology Group (GRACUS)
Host: Prof. Jose Beltrán Jiménez
Collaboration on EFT constructions of dark sector models and perturbative stability.

SISSA June 2025
Astroparticle and Gravitational Physics Group
Host: Prof. Enrico Barausse
Collaboration on k -essence and astrophysical screening mechanisms.

Babeş-Bolyai University October 2024
Department of Physics
Host: Prof. Tiberiu Harko
Collaboration on geometric modifications of gravity and cosmological constraints.

University of Science and Technology of China April–May 2023
Department of Astronomy
Host: Prof. Yi-fu Cai
Collaboration on effective field theory approaches to strong coupling in modified gravity.

University of Tartu August–October 2021
Laboratory of Theoretical Physics

Host: Prof. Tomi S. Koivisto
 Collaboration on Lagrangian formulations of relativistic interacting fluids.

Talks and Presentations

Invited international conference talks

Lagrangian models of interacting dark matter and dark energy Jul 2026
[GGI Workshop: Exploring New Frontiers in Cosmology](#), Galileo Galilei Institute, Italy

Pure momentum transfer from non-minimal couplings Jun 2026
 Mini Workshop on Cosmology with Elastic Dark Interactions: Current Progress and Open Challenges, Universidad Politécnica de Madrid, Spain

Interacting dark sector from entropy couplings Sep 2025
[Tensions in Cosmology, Corfu2025](#), Corfu Summer Institute, Greece

Geometric modifications of gravity Apr 2023
[International Conference of Deep Space Sciences](#), USTC, China

Invited seminar talks

Thermodynamic interpretations of extended dark sector models in cosmology Oct 2026
[Institute of Theoretical Physics Seminar](#), University of Wrocław, Poland

Intrinsic entropy perturbations in cosmology Jun 2026
[CMB/LSS Meeting, Kavli Institute for Cosmology](#), University of Cambridge, UK

Interacting dark sector with intrinsic entropy couplings Mar 2026
[CosmoVerse Journal Club Seminar](#), Online

Interacting dark sector imprints on matter clustering Feb 2026
[High-Energy Physics Seminar](#), University of Southampton, UK

Pure momentum transfer from dark entropy couplings Jan 2026
[London Relativity and Cosmology Seminar](#), Queen Mary University, UK

Intrinsic entropy couplings in the dark sector Oct 2025
[Gravitation and Cosmology Seminar](#), University of Sheffield, UK

Entropy interactions between dark matter and dark energy Oct 2025
[Gravitation and Cosmology Group Seminar](#), Universidad de Salamanca, Spain

Geometric k -essence from nonmetricity and late-time cosmology Jun 2025
[Astroparticle and Gravitational Physics Seminar](#), SISSA, Italy

Interacting fluids in cosmology: Lagrangian approaches and extensions Nov 2024
[Universe and Particles Laboratory Seminar](#), University of Montpellier, France

Advanced dynamical systems methods in cosmology Oct 2024
 Physics Department Seminar, Babeş-Bolyai University, Romania

Second-order modifications of gravity Jun 2023
[DAMTP General Relativity Seminar](#), University of Cambridge, UK

Studying $f(T)$ and $f(Q)$ gravity with dynamical systems in cosmology May 2023
[Particle Cosmology Group Meeting](#), USTC, China

Cosmological dynamical systems in modified gravity Jun 2022
[London–Oldenburg Relativity Seminar](#), Online

Cosmological dynamical systems in modified gravity May 2022
[HEP–Cosmo–Astro Seminar](#), Online

Contributed international conference talks

Viable interactions in the dark sector Jun 2026
[31st International Symposium on Particles, Strings, and Cosmology \(PASCOS\)](#), University of Sheffield, UK

Entropy interactions in the dark sector Apr 2026
[Iberian Cosmology Meetings \(IberiCos 2026\)](#), University of Granada, Spain

k-essence and cosmic expansion from integrable vectorial nonmetricity May 2025
[13th Bolyai–Gauss–Lobachevsky Conference on Non-Euclidean Geometry in Modern Physics and Mathematics](#), Morocco

Poster presentations

Cosmological fluids in modified gravity Oct 2024
[COSMO’24](#), Kyoto University, Japan

Modifying gravity by breaking fundamental symmetries Jul 2024
[CosmoVerse@Krakow](#), Jagiellonian University, Poland

Metric-affine gravity and its modifications Jan 2023
[UK Cosmology Meeting](#), University of Sussex, UK

Teaching Experience

Lecturing

- *Mathematics for General Relativity* (Department of Mathematics) 2023 & 2024
 Covering & delivering lectures. Course leader: Prof. C. Boehmer

Teaching

- *Computing for Mathematical Physics* (Department of Physics) 2022 & 2023
 Senior Postgraduate Teaching Assistant
- *An Introduction to Mathematica* (Department of Mathematics) 2021
 Postgraduate Teaching Assistant

UCL Arena Teaching Development Programme 2021–2022

Supervision

- BSc 3rd year project (UCL Department of Mathematics) 2025 - 2026
 Co-supervision of 3rd year dissertation: “*Gravitational waves in $f(G)$ gravity*”
- [MAPS Summer Research Internship](#) (UCL Department of Mathematics) 2024
 Co-supervision of research project on Lie groups in theoretical physics; student subsequently progressed to a PhD within the group.

Professional Experience

Senior Algorithm Engineer, MBDA, Bristol, UK 2018 - 2019
 Designing algorithms using Python, performing statistical analyses, and using machine learning and AI techniques.

Service and Outreach	<i>In2Research Mentor</i> at <i>In2scienceUK</i>	2024 - 2025
	Mentoring a student from an underrepresented background towards postgraduate research.	
	<i>Equality, Diversity and Inclusion (EDI) committee</i>	2021 - present
	Department of Mathematics, University College London Co-designed postgraduate EDI survey; findings informed new departmental mentoring initiatives	
	<i>Blackett Lab Family (BLF)</i>	2022 - present
Computational Expertise	Invited speaker at <i>Representing Physics 2024</i>	
	Organiser of <i>Philosophy and space-time reading group</i>	2024 - present
	Department of Mathematics, University College London	
	Reviewer for <i>Classical and Quantum Gravity</i> , <i>Nuclear Physics B</i> , <i>JCAP</i> , <i>Journal of Mathematical Physics</i> , & <i>Physics Letters B</i>	2023 - present
	<i>CLASS</i> , <i>Cobaya</i> , <i>Mathematica/xAct</i> , <i>Python</i> , <i>C</i> , <i>Bayesian inference</i> and <i>MCMC methods</i> .	